Projected force, global Doob signed Gram, adaptive collar, and quotient boundary

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1. Purpose and scope

R-140 proved that the production exponents make the positive triangular analysis summable, but its current-only constants were too large for the conditional source headroom. It therefore left a precise structural task: keep the complete R-102 product covariance, primitive trace, R-125 future variance, R-063 forest, and mean/low coordinate in one signed response before taking absolute values.

The present checkpoint advances that task in four ways. First, it writes the actual finite-chart projected force of the complete action owner and shows that the covariance-normal variance force cancels rather than being charged again. Second, it gives a canonical three-coordinate signature for the complete trace/mean/current/variance owner after exhaustive R-125 recombination. Third, it conjugates the whole finite-chart Hessian to a global Doob source coordinate without deleting off-diagonal reveal blocks and states the exact source-null quotient and low-kernel acceptance criteria. Fourth, it proves the exact positive-analysis collar shift, computes the $C = 8$ and $C = 10$ diagnostics, and identifies the coherent finite-band price of moving the collar.

The projected-force formula is an explicit assembly of R-126–R-129, not a claim that a new physical force has been bounded. The quotient criterion is an application of the R-118/R-129/R-140 Douglas machinery, not a second Douglas theorem. The new durable coordinate is the complete canonical signature and its global Doob signed-Gram placement.

Everything is at finite cutoff, positive floor, and a fixed temporally faithful chart. The two-sided production factorisation, its uniform upper Loewner constant, the production common trace feature, finite-collar blocks, low compatibility, and a positive graph gap remain unproved. No matching, absolute anchor, A13 gate, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson estimate, removal, interacting measure, Sector-A closure, or tier promotion follows.

2. Exact finite-chart production projected force

Let π be a finite temporally faithful source chart and

$$Z_h = S_\pi \xi + L_\pi h. \tag{2.1}$$

The complete action owner is

$$\mathcal{A}_\pi(h) = \sum_\alpha \mathbb{E}\,\mathcal{P}_{\mathrm{comp},\alpha}(Z_h) + \frac{9}{20}\|h\|^2_{\mathcal{S}_\pi} + \frac{3}{20}\mathbb{E}\|Z_{K,h}\|_6^6, \tag{2.2}$$

where every exhaustive low/injected scalar owner is included once and

$$\mathcal{P}_{\mathrm{comp}} = \frac{1}{2}\|\Phi\|^2 - \frac{1}{2}\Theta = -\frac{1}{2}\mathcal{T}. \tag{2.3}$$

If $G_{\mathcal{T}}$ denotes the R-126 Euler force of the complete trace-excess owner, then

$$G_{\mathrm{comp}} = -\frac{1}{2} G_{\mathcal{T}}. \tag{2.4}$$

Equivalently, if $G_{\mathcal{V}}$ is the future-variance coordinate and $G_{\mathrm{CN}} = (G_{\mathcal{V}} - G_{\mathcal{T}})/2$, then

$$G_{\mathrm{CN}} - \frac{1}{2} G_{\mathcal{V}} = -\frac{1}{2} G_{\mathcal{T}}. \tag{2.5}$$

Thus the variance force disappears when the covariance-normal coordinate is returned to the smaller action owner. Appending it again would duplicate an owner.

Put

$$G_6(Z) = \frac{9}{10} |Z|^4 Z, \quad G_{\mathrm{end}} = G_6 - \frac{1}{2} \sum_{\alpha} G_{\mathcal{T},\alpha}. \tag{2.6}$$

The legal projected force is

$$\boxed{(g_\pi^{\mathrm{prod}}(h))_b = \frac{9}{10} h_b + \mathbb{E}\left[S_b^* G_{\mathrm{end}}(Z_h) \mid \mathcal{F}_{t_{b-1}} \right].} \tag{2.7}$$

The conditional expectation is the orthogonal projection to the predictable source block. Differentiating on the finite chart gives

$$\boxed{D g_\pi^{\mathrm{prod}}(h) = \frac{9}{10} \mathrm{I} + P_{\mathrm{pred}} L_\pi^* (Q_{\mathrm{comp}} + Q_6) L_\pi P_{\mathrm{pred}}.} \tag{2.8}$$

It is self-adjoint. The physical pullback in (2.8) kills $\operatorname{Ker} L_\pi$; the explicit source cost $9\mathrm{I}/10$ does not. Under a representation-preserving refinement ι,

$$g_\pi(h) = \iota^* g_{\pi'}(\iota h), \quad H_\pi(h) = \iota^* H_{\pi'}(\iota h) \iota. \tag{2.9}$$

Forward and legal reverse are therefore the two adjoint faces of one Hessian owner. The reverse has constant one and is not a second charge.

The executable certificate differentiates (2.2) on two unrelated polynomial charts and checks (2.7) by exact Rademacher conditional expectation. These checks establish the finite-chart algebra only; they give no uniform norm.

3. Canonical complete signature owner

At a fixed complete endpoint q, let

$$\Phi_q = \mathbb{E}_f J_q, \quad R_q = J_q - \Phi_q, \quad V_q = \mathbb{E}_f \|R_q\|^2. \tag{3.1}$$

Choose the *actual common covariance/trace factor* U_q in one pinned feature space, with

$$\mathbb{E}_f \|U_q\|^2 = \Theta_q. \tag{3.2}$$

The word "actual" is essential. An arbitrary vector with the same norm does not determine $\langle U_N, U_p \rangle$ and cannot determine the mixed Gram; that shortcut is excluded by the R-131 diagonal-Gram information boundary.

Define

$$\mathbf{Z}_q = (U_q, J_q, R_q), \quad \mathrm{S} = \operatorname{diag}(\mathrm{I}, -\mathrm{I}, \mathrm{I}). \tag{3.3}$$

Then the pointwise complete signature identity is

$$\boxed{\mathbb{E}_f \langle \mathbf{Z}_q, \mathrm{S} \mathbf{Z}_q \rangle = \Theta_q - \mathbb{E}_f \|J_q\|^2 + V_q = \Theta_q - \|\Phi_q\|^2.} \tag{3.4}$$

Writing $\Phi = b + Y$ relative to the previous reveal gives

$$\mathbb{E}_{r-1}\langle \mathbf{Z}_q, \mathrm{S}\mathbf{Z}_q\rangle = D_{0,q} - \|b_q\|^2. \tag{3.5}$$

Only after the exhaustive contraction-closed output sum, fixed-root ownership, and same-root visit telescope of R-125 may the alternate forest coordinate be inserted:

$$\sum_{r,m} w^C_{r,m}\mathbb{E}\Delta_{N,p}\langle \mathbf{Z}, \mathrm{S}\mathbf{Z}\rangle = \sum_{r,m} w^C_{r,m}\mathbb{E}\Delta_{N,p}(V - \mathrm{Forest}^{\mathrm{ad}}_{063}). \tag{3.6}$$

Equation (3.6) is a complete aggregate identity, not an arbitrary cell-by-cell equality.

Let the common value in (3.6) be S^{full}_C. The R-139 future action and the returned mean/low owner obey

$$\boxed{\mathfrak{I}^C_{\mathrm{fut}} + \frac{1}{2}\sum_{r,m} w^C_{r,m}\mathbb{E}\Delta_{N,p}\|b\|^2 = -\frac{1}{2}S^{\mathrm{full}}_C.} \tag{3.7}$$

The primitive trace appears once, the raw current once, the future variance once, and the mean is returned once. The forest is an alternate coordinate with multiplicity zero; it is never appended as an extra reserve.

The exact audit fixture

$$\|b\|^2 = 9, \quad \mathbb{E}\|Y\|^2 = 4, \quad \Theta = 6, \quad V = 5 \tag{3.8}$$

has $\|\Phi\|^2 = 13$, $\mathbb{E}\|J\|^2 = 18$, Forest $= 12$, and signature -7. Its future owner is -1, while returning $\|b\|^2/2$ gives $7/2 = -(-7)/2 = \mathcal{P}_{\mathrm{comp}}$. Omitting variance gives -12 and duplicating the forest gives -19, so both mutations are detected.

4. Conditional two-sided signed mixed Gram

Let $\widehat{x}_r = (\ell_r, x_r)$ contain the once-owned low coordinate and all still-future source innovations at reveal r. Suppose predictable common- feature operators satisfy

$$W_C^{1/2}(\mathbf{Z}_N - \mathbf{Z}_p) = A^-_r\widehat{x}_r, \quad W_C^{1/2}(\mathbf{Z}_N + \mathbf{Z}_p) = A^+_r\widehat{x}_r. \tag{4.1}$$

This is the missing production factorisation: it must contain the complete R-102 product residual and the canonical trace lift of Section 3.

Theorem 4.1. Under (4.1), define

$$Q^C_r = \frac{1}{2}\left[(A^+_r)^*\mathrm{S}A^-_r + (A^-_r)^*\mathrm{S}A^+_r\right]. \tag{4.2}$$

Then Q^C_r is self-adjoint and

$$S^{\mathrm{full}}_C = \sum_r \mathbb{E}\langle \widehat{x}_r, Q^C_r\widehat{x}_r\rangle. \tag{4.3}$$

If, for a positive source/low metric G_r,

$$Q^C_r \preceq \Lambda^2_C G_r, \quad \sum_r \mathbb{E}\langle \widehat{x}_r, G_r\widehat{x}_r\rangle \le X_{\mathrm{src}} + X_{\mathrm{low}}, \tag{4.4}$$

then

$$\mathfrak{I}^C_{\mathrm{fut}} + \text{mean/low return} \ge -\frac{\Lambda^2_C}{2}(X_{\mathrm{src}} + X_{\mathrm{low}}). \tag{4.5}$$

Proof. Polarisation gives

$$\Delta\langle \mathbf{Z}, \mathrm{S}\mathbf{Z}\rangle = \mathrm{Re}\langle \mathbf{Z}_N + \mathbf{Z}_p, \mathrm{S}(\mathbf{Z}_N - \mathbf{Z}_p)\rangle.$$

Substitute (4.1); the skew-adjoint part vanishes on the quadratic diagonal, which yields (4.2)–(4.3). Apply the one-sided Loewner bound and then (3.7). There is no additional factor 1/2 inside Q^C_r: the only action half is already displayed in (3.7). ☐

Only the upper Loewner edge is adverse. For example, $\mathrm{diag}(-100, 1/20) \preceq (1/20)\mathrm{I}$ even though its absolute operator norm is 100. Replacing (4.4) by an absolute norm can therefore destroy the cancellation that R-140 requires.

5. Global Doob-coordinate conjugation

For predictable source blocks $h_k \in L^2(\mathcal{F}_{k-1}; H_k)$ define

$$\mathcal{D}h = \big((P_\ell h_k)_k, (d_r h_k)_{\ell<r<k}\big). \tag{5.1}$$

Martingale Pythagoras gives the exact source isometry

$$\|\mathcal{D}h\|^2 = \sum_k \mathbb{E}\|h_k\|^2. \tag{5.2}$$

Let H be the complete bounded self-adjoint finite-chart Hessian form. On $\operatorname{Ran}\mathcal{D}$ put

$$Q_{\text{glob}} = \mathcal{D}H\mathcal{D}^*, \tag{5.3}$$

and extend it by zero on the orthogonal complement. Then

$$\boxed{\langle h, Hh\rangle = \langle \mathcal{D}h, Q_{\text{glob}}\mathcal{D}h\rangle.} \tag{5.4}$$

Equation (5.4) is a conjugation, not a reveal-diagonalisation theorem. A random terminal Hessian need not commute with martingale projections, so Q_{glob} generally retains $r \neq s$ reveal blocks. The R-102 coefficient-current product covariance also remains inside these blocks. Deleting them is an unsupported positive-cell replacement. Forward, legal reverse, balanced, and low terms are blocks of this one global self-adjoint form.

This global route is weaker in appearance but safer than demanding a separate per-reveal factorisation. A uniform upper Loewner estimate for Q_{glob} would suffice. R-141 constructs the coordinate and firewall; it does not prove that estimate.

6. Exact source-null quotient and low-kernel tests

Let $S : \mathcal{U} \to \mathcal{X}$ be source synthesis and split

$$\mathcal{U} = \operatorname{Ker} S \oplus (\operatorname{Ker} S)^\perp, \quad A_1 = (S^*S)|_{(\operatorname{Ker} S)^\perp} > 0. \tag{6.1}$$

Write the complete signed Hessian in blocks

$$H_C = \begin{pmatrix} H_{00} & H_{01} \\ H_{10} & H_{11} \end{pmatrix}. \tag{6.2}$$

Theorem 6.1. In finite dimension, a finite Λ_C with

$$H_C \preceq \Lambda_C^2 S^* S \tag{6.3}$$

exists exactly when

$$D_0 = -H_{00} \succeq 0, \quad \operatorname{Ran} H_{01} \subseteq \operatorname{Ran} D_0^{1/2}, \tag{6.4}$$

and

$$H_{11} + H_{10}D_0^\dagger H_{01} \preceq \Lambda_C^2 A_1. \tag{6.5}$$

In Hilbert space, (6.4)–(6.5) use the closed-range/reduced-solution form.

Proof. Subtract (6.2) from the right side of (6.3) and apply the positive semidefinite block-matrix criterion to

$$\begin{pmatrix} D_0 & -H_{01} \\ -H_{10} & \Lambda_C^2 A_1 - H_{11} \end{pmatrix}. \tag{6.6}$$

The range condition and Schur complement are precisely (6.4)–(6.5). ☐

If $H_{00} = 0$, (6.4) forces $H_{01} = 0$. A clean stronger structural form is

$$H_C = S^* \widetilde{H}_C S, \quad \widetilde{H}_C \preceq \Lambda_C^2 \mathrm{I}. \tag{6.7}$$

The executable quotient fixture adds a positive rank-one form on $\operatorname{Ker} S$; the source metric stays zero there, so no finite constant can dominate the mutation.

For the low block let $D = L^* L \succeq 0$. The constructive common-feature condition

$$K^* Z = L^* R \tag{6.8}$$

implies $P_{\operatorname{Ker} D} K^* Z = 0$ and hence exact low compatibility. More generally, on the production graph let

$$N = P_{\operatorname{Ker} D} K^*, \quad \Lambda_0 = P_{\operatorname{Ker} D} \Lambda, \quad C_{\ker} = N^* \Lambda_0 + \Lambda_0^* N \preceq \kappa_0 \mathrm{I}. \tag{6.9}$$

The kernel cross then costs at most κ_0 in the source diagonal. With source diagonal e, companion diagonal f, low Schur cost σ, base cross a_0, and residual tail τ_C, the exact conditional margin is

$$\boxed{\mu_{\mathrm{src}} = e - \Lambda_C^2 - \sigma - \kappa_0 - \frac{(a_0 + \tau_C)^2}{4(f - \sigma)}.} \tag{6.10}$$

Exact compatibility has $\kappa_0 = 0$. Radial Taylor contributes only $\mu_{\mathrm{src}}/2$ to the action.

The fixture $D = \operatorname{diag}(1, 0)$ with a unit kernel cross has graph form $1 - 2t$; at $t = 3/4$ it is $-1/2$. Its exact penalty is $\kappa_0 = 3/2$. Thus $D \succeq 0$ alone does not imply compatibility.

7. Exact positive-analysis collar shift

For the R-140 positive triangular analysis operator

$$(A_r^C x)_m = \mathbf{1}_{\{m \geq r + C\}} 2^{\gamma(m - r - C)} \sum_{k > r} T_{r,m,k} x_k, \tag{7.1}$$

the collar dependence is exact.

Theorem 7.1. For every integer $q \geq 0$,

$$\boxed{A_r^{C+q} = 2^{-\gamma q} P_{\{m \geq r + C + q\}} A_r^C.} \tag{7.2}$$

Consequently

$$\|A_r^{C+q}\| \leq 2^{-\gamma q} \|A_r^C\|, \quad \Lambda_{C+q}^2 \leq 2^{-2\gamma q} \Lambda_C^2. \tag{7.3}$$

Proof. On the surviving rows the weight changes by $2^{-\gamma q}$; the projection deletes all other rows. ☐

This theorem applies only to a positive analysis norm. It does not imply a Loewner monotonicity for an indefinite signed Q_C: removing a negative shell can increase the upper eigenvalue.

At

$$\beta = \frac{7}{5}, \quad s = \frac{2}{3}, \quad \gamma = \frac{7}{12}, \quad P = 4 + 10^{-12}, \tag{7.4}$$

the exact R-140 scalar sum gives

(7.5)

C	H_C	$\frac{3H_C}{40P}$
7	10.778126471501523	0.202089871340603
8	4.631138871013502	0.0868338538314814
9	1.971947657656302	0.0369740185810464
10	0.833374378872430	0.0156257696038541

Under the diagnostic normalization $C_* = 1$, $C = 8$ is the first integer below the zero-low/tail headroom $0.1004343434\ldots$, with

$$C_* < 1.07546575053067. \tag{7.6}$$

For the illustrative smaller headroom 0.0159126895, $C = 10$ is first, with

$$C_* < 1.00913922178312. \tag{7.7}$$

The debt scales as C_*^2, not C_*. These are diagnostics because no production value of C_* has been proved.

8. What the larger collar costs

Changing $C = 5$ to $C = 8$ moves offsets $5, 6, 7$ into a finite collar. Changing to $C = 10$ moves all five offsets $5, \ldots, 9$. They do not disappear. For the oriented complete collar blocks T_{mr} put

$$R_B = \sup_m \sum_r \|T_{mr}\|, \quad S_B = \sup_r \sum_m \|T_{mr}\|. \tag{8.1}$$

Operator-valued Schur gives

$$\boxed{\|T_B\| \le \sqrt{R_B S_B}.} \tag{8.2}$$

If the moved layers obey $\|T_{r+q,r}\| \le b_q$, then

$$\|T_B\| \le \sum_{q \in B} b_q, \quad a_B^{\text{full}} \le 2 \sum_{q \in B} b_q. \tag{8.3}$$

The factor two converts the oriented block to the full-cross Hessian convention. Its adjoint is the legal reverse already present in the same owner; adding it again would double count.

The exact cyclic-band fixture $b_5 = 1/6$, $b_6 = 1/4$, $b_7 = 1/3$ has row and column sums $3/4$, and the constant vector attains $\|T_B\| = 3/4$. Therefore the square-root-of-squares shortcut is false for coherent layers.

Using only the R-130 local Cartan diagnostic,

$$c_{\text{Cartan}}^{\text{or}} = \frac{67}{1200P}, \quad C_{\text{br}} = 17.1842828361\ldots, \tag{8.4}$$

one layer would have the crude full-cross size

$$2c_{\text{Cartan}}^{\text{or}} C_{\text{br}} = 0.479727895841\ldots. \tag{8.5}$$

Paying three layers by a triangle gives $1.43918368752\ldots$, already above the old diagnostic $K_0 = 0.920932333908\ldots$. This is not a production counterexample: it shows that a layerwise absolute triangle is too crude and that one signed finite-band/mixed-replica Gram must be evaluated.

An adaptive $C = 8/10$ choice is legal only when it is $\mathcal{F}_{r-1}$-measurable and constant across every insertion endpoint in the R-139 telescope. Otherwise a moving-mask variation term returns. A $C = 10$ branch also needs separate production bounds for offsets $8, 9$; a three-layer theorem is insufficient.

9. Signed correlation is indispensable

Let

$$E = 0.8857272727272757, \quad F = 0.27, \quad a = 0.9209323339075279, \quad d = 1, \quad b = \frac{1}{\sqrt{50}}. \tag{9.1}$$

Consider

$$M_+ = \begin{pmatrix} E & a/2 & b \\ a/2 & F & b \\ b & b & 1 \end{pmatrix}, \quad M_- = \begin{pmatrix} E & a/2 & b \\ a/2 & F & -b \\ b & -b & 1 \end{pmatrix}. \tag{9.2}$$

The matrices have identical diagonal and magnitude data. Eliminating the low coordinate gives

$$\Delta_+ = (E - 1/50)(F - 1/50) - (a/2 - 1/50)^2 = 0.0224213739509\ldots, \tag{9.3}$$

$$\Delta_- = (E - 1/50)(F - 1/50) - (a/2 + 1/50)^2 = -0.0144159194054\ldots. \tag{9.4}$$

Their least eigenvalues are respectively $0.02039620\ldots$ and $-0.01232585\ldots$. Thus the same $E, F, a, d, |b|, |c|$ data pass or fail solely through the signed correlation. Magnitude-only budgets cannot certify the graph gap.

This fixture specializes the existing R-127/R-131 information boundary. The low-kernel fixture specializes R-127/R-140, and failure of signed collar monotonicity specializes the R-139/R-140 cancellation firewall. No new negative-result ID is created; the route decisions are preserved in the exploration ledger.

10. Combined conditional acceptance theorem

The preceding pieces can now be stated without hidden duplication.

Theorem 10.1. Fix a finite production chart. Assume:

1. the actual common covariance/trace features and exhaustive R-125 owner recombination give (3.4)–(3.7);
2. the complete action secant has either the predictable factorisation (4.1) or a global Doob-coordinate upper Loewner bound, with source debt Λ_C^2;
3. every moved collar layer is retained in the same signed Hessian and the adaptive collar is predictable and endpoint-fixed;
4. the source-null conditions (6.4)–(6.5) and the low-kernel estimate (6.9) hold; and
5. the production values satisfy $f > \sigma$ and $\mu_{\rm src} > 0$ in (6.10).

Then the complete finite-chart Hessian has source-graph gap $\mu_{\rm src}$, and the radial action has source contribution at least $\mu_{\rm src}/2$.

Proof. The signed owner is represented once by (4.3) or (5.4). The one-sided source debt is removed from the source diagonal. The source- null and low-kernel blocks are eliminated by Theorem 6.1 and (6.9), while the remaining two-channel cross is completed sharply. This gives (6.10). Integrating the Hessian along the radial Taylor path supplies the factor 1/2. □

The theorem is conditional because assumptions 1–4 have not been proved with production-uniform constants. The diagnostics in Section 7 do not verify assumption 5.

11. Evidence map and successor route

The proof map is now narrower than after R-140:

1. R-126/R-128 supply the exact physical force and pullback; Section 2 assembles the actual complete action force without variance duplication.
2. R-125/R-063 and Section 3 supply one canonical complete signature after exhaustive recombination.

3. R-102's product covariance must be retained inside the two-sided or global signed Gram of Sections 4–5; an x_r-only coefficient factorisation is known to fail.
4. R-118/R-129/R-140 reduce acceptance to the exact source-null and low- kernel tests of Section 6.
5. The positive tail can be reduced by $C = 8$ or $C = 10$, but Section 8 requires the corresponding coherent three- or five-layer signed band.
6. The decisive successor is therefore to construct the actual common- heat paired-replica feature maps for the entire R-102 Hessian, compute the global signed Gram before absolute values, and evaluate the full source, companion, finite-band, and low matrix. If it fails, its adverse signed eigenvector identifies the saturating owner. If it passes, matching and the absolute anchor become the next gates.

This checkpoint stops here. It has replaced several looping scalar routes by one exact operator target, but it has not closed T-050 or Sector A.

12. Devil's-advocate audit

1. **Objection:** any U_q with norm $\Theta_q^{1/2}$ defines the signature Gram. **UPHELD as an overclaim.** Norms do not determine mixed endpoint correlations; the actual pinned common feature is required.
2. **Objection:** $V -$ Forest holds cellwise. **UPHELD as an overclaim.** It is licensed only after the complete R-125 contraction/root/visit recombination.
3. **Objection:** Doob isometry makes the Hessian reveal-diagonal. **UPHELD as an overclaim.** It diagonalises the source norm only; the global conjugated Hessian retains off-diagonal reveal blocks.
4. **Objection:** the legal reverse costs another copy. **DISMISSED.** It is the adjoint block of the same self-adjoint Hessian.
5. **Objection:** the signature Gram needs a second factor 1/2. **DISMISSED.** Polarisation fixes (4.2); the action half occurs once in (3.7).
6. **Objection:** a large negative eigenvalue enlarges the adverse debt. **DISMISSED.** Only the upper Loewner edge enters the lower action bound.
7. **Objection:** positive semidefinite D removes the low kernel. **DISMISSED.** The $1 - 2t$ fixture is negative at $t = 3/4$.
8. **Objection:** increasing the collar monotonically improves any signed Gram. **UPHELD as an overclaim.** The shift theorem is for the positive analysis operator only.
9. **Objection:** $C = 10$ costs the same three layers as $C = 8$. **DISMISSED.** It moves offsets $5, \dots, 9$ and needs a five-layer bound.
10. **Objection:** the $C = 8/C = 10$ numbers prove headroom. **UPHELD as an overclaim.** They assume diagnostic $C_* = 1$ and omit the production signed finite band and low blocks.
11. **Objection:** magnitude-only Schur data suffice. **DISMISSED.** The matrices (9.2) have identical magnitude data and opposite positivity verdicts.
12. **Objection:** a positive conditional μ_{src} would close Sector A. **UPHELD as an overclaim.** Matching, the absolute anchor, Nelson, and removal would still remain.

13. Result footer and reproduction

Primary certificate:

```
E:\Dev\TECT.venv\Scripts\python.exe `
  codes/foundations/a13_classii_projected_force_global_doob_signed_gram_`
  adaptive_collar_quotient_boundary.py
```

Independent certificate:

```
E:\Dev\TECT.venv\Scripts\python.exe `
  codes/foundations/a13_classii_projected_force_global_doob_signed_gram_`
  adaptive_collar_quotient_boundary_independent.py
```

Integrated verification:

```
E:\Dev\TECT.venv\Scripts\python.exe `
  codes/foundations/a13_classii_projected_force_global_doob_signed_gram_`
  adaptive_collar_quotient_boundary_verify.py
```

```
Result ID: A13-CLASSII-PROJECTED-FORCE-GLOBAL-DOOB-SIGNED-GRAM-
  ADAPTIVE-COLLAR-QUOTIENT-BOUNDARY.
Ledger ID: R-141.
Precise statement: finite-chart projected force (2.7)-(2.9); canonical
  complete signature (3.4)-(3.7); conditional signed Gram Theorem 4.1;
  global Doob conjugation (5.4); source-null Theorem 6.1; low-kernel gap
  (6.10); positive collar Theorem 7.1; conditional Theorem 10.1.
Scope: finite cutoff, positive floor, fixed temporally faithful chart;
  complete R-125 recombination; displayed common-feature, factorisation,
  quotient, low-kernel, collar, and production-constant hypotheses only.
Dependencies: A1; R-063, R-102, R-118, R-125, R-126, R-127, R-128,
  R-129, R-130, R-131, R-132, R-136, R-139, R-140.
Evidence grade: ANALYTIC, EXACT, CONDITIONAL, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a13_classii_projected_force_global_doob_signed_gram_
  adaptive_collar_quotient_boundary_verify.py
Expected output: R-141 integrated PASS; primary and independent rows,
  deterministic note/PDF, manifest, exploration, authority, and public
  surface gates pass.
Falsification gate: any failed force cancellation, conditional projection,
  signature multiplicity, mixed polarization, global Doob conjugation,
  quotient/range, low-kernel, collar shift/threshold/band, signed-correlation,
  authority/hash, PDF, or public-surface check.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no production common-feature factorisation, uniform
  upper Loewner constant, finite-band or low compatibility, positive graph
  gap, matching, anchor, A13 gate, OVERLAP_src, Nelson estimate, removal,
  interacting measure, Sector-A closure, or T5-T7 claim follows.
Next required action: after this requested stop, construct or falsify the
  complete common-heat paired-replica/global-Doob signed Gram, retain the
  coherent C=8 or C=10 finite band and low kernel, and evaluate (6.10) before
  matching and the absolute anchor.
```